

Joshua Opria

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Summary

Senior Software Engineer with 7+ years building full-stack and distributed systems, now focused on applied AI/ML and high-scale cloud platforms. Proven track record designing LLM-powered pipelines, document intelligence systems, and event-driven data platforms in production environments.

Independent researcher and open-source contributor, with hands-on experience building end-to-end AI products and scalable real-time systems. Lead projects, mentor engineers, and focus on system design, evaluation, and the long-term health of what I build.

Skills

- AI / ML: PyTorch, TensorFlow, LLMs, Transformers, Hugging Face, NLP, RAG, Information Extraction, Azure ML
- Data & Pipelines: Kafka, Azure Service Bus, Event-Driven Architectures, Pydantic, ETL Pipelines
- Cloud / Infra: AWS, Azure, Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, DataDog, Helm
- Languages: Python, TypeScript, Java, Elixir, SQL, C++, Ruby
- Frameworks: React, Node.js, Django, Spring Boot, Rails, Prisma
- Other: Algorithms, Experimentation, A/B Testing, API Design, CI/CD, Mentorship

Work Experience

CoverMyMeds | Columbus, OH - Software Engineer II | May 2022 – Present

- Architected LLM powered document intelligence pipelines for prior authorization workflows using OCR, Azure AI, and structured schema validation (Pydantic)
- Designing template free IDP systems to extract structured data from PDFs, EHR forms, and clinical documents
- Built event-driven processing pipelines (Kafka / Service Bus) to normalize and transform healthcare data at scale
- Implementing validation and evaluation layers to ensure accuracy, completeness, and data consistency across extracted outputs
- Delivered internal talk on AI Agents & AI Factories, shaping architectural direction of AI systems
- Architected distributed Elixir + Azure Service Bus system improving sponsorship accuracy by 50% (+\$7.8M annual revenue)
- Led development of APIs handling 5K+ req/min and frontend systems serving 140K+ req/min
- Built SOX-compliant cloud infrastructure using Terraform and GitHub Actions
- Mentored 7+ engineers on system design and cloud best practices

Nationwide Insurance | Columbus, OH - Technical Lead | Dec 2020 – May 2022

- Designed and deployed GraphQL APIs (AWS AppSync) consolidating multiple data sources; reduced response times by 15%.
- Mentored 9 peers on cloud technologies, system design, and architectural patterns - increasing project throughput by 20%.
- Served as technical liaison with business partners to align architecture with measurable business outcomes.

Nationwide Insurance | Columbus, OH - Senior Software Engineer | Mar 2020 – Dec 2020

- Implemented S3-based document storage and CI/CD pipelines (Concourse, Jenkins).
- Delivered Spring Boot APIs handling 20 K+ req/min and custom Node.js OAS validator improving API quality by 70%.

Nationwide Insurance | Columbus, OH - Software Engineer | Jun 2019 – Mar 2020

- Created internal Angular SPA for 200+ APIs, supporting developer access control.

Nationwide Insurance | Columbus, OH - Software Developer Contractor | Dec 2018 – Jun 2019

- Created Angular SPA for 200+ APIs and built 7 Apigee services for auth workflows with automated Concourse deployments.

Scenicio | Bowling Green, OH - Software Developer | Aug 2018 – Dec 2018

- Developed an AR mobile app (Kotlin, Android/iOS) integrating real-world data from government APIs.

Nationwide Insurance | Columbus, OH - Software Developer Intern | Jun 2018 – Aug 2018

- Containerized and deployed Java Spring Boot services in AWS Kubernetes (EKS).

Projects

STRUM - Audio-to-Chart ML Pipeline

arxiv.org/abs/2605.12135 · github.com/opria123/strum

- Authored a self-published research paper, introducing an end-to-end multi-instrument pipeline that converts raw audio into playable Clone Hero / YARG charts for drums, guitar, bass, vocals, and keys without oracle metadata.
- Architected a two-stage CRNN drum transcriber and 6-model ensemble classifier across 8 lanes, reaching 93.9% onset F1 and 85.2% lane-classification F1 on a held-out pool of 3,299 community-authored pro-drum charts.
- Released a 29-song in-envelope benchmark with drums 83.8% F1, bass 69.4%, guitar 65.1%, vocals 53.9% end-to-end vs. human-authored ground truth.
- Stack: Python, PyTorch, Demucs, Whisper, Spotify Basic Pitch, librosa, Hydra, Weights & Biases; trained on NVIDIA DGX Spark

OCTAVE - Desktop Chart Editor & Auto-Charter

octavestudio.tools · github.com/opria123/octave

- Shipped a cross-platform (Windows / macOS / Linux) desktop app that wraps STRUM in a one-click "audio to playable chart" workflow with a 2D piano-roll editor and live 3D highway preview.
- Bundled a self-contained Python 3.11 runtime, PyTorch, Demucs, and Whisper inside the Electron app so end users get a 100% local.
- Built the 8-instrument with full round-trip support for .mid and .chart, preserving tempo maps, sections, lyrics, and metadata.
- Implemented the live 3D highway in Three.js / React Three Fiber over a Zustand, Zundo undo/redo store, keeping the 2D editor, 3D scene, and multi-stem audio mixer in lockstep during playback.
- Stack: Electron, React, TypeScript, Three.js / React Three Fiber, Zustand, Zundo

YARG Online Multiplayer

github.com/opria123/YARG.Networking

- Designed and built the first online multiplayer implementation for YARG, an open-source rhythm game.
- Implemented connectivity on LiteNetLib with NAT punch-through and relay-server fallback, allowing peers behind restrictive routers to connect without manual port forwarding.
- Shipped a Dockerized dedicated server, deployed via docker-compose and Fly.io via a GitHub Actions CI pipeline
- Stack: C# / .NET 8, Unity, Docker, Fly.io

Education and Certifications

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| • MSc. Computer Science, Artificial Intelligence and Machine Learning , WGU | Oct 2025 |
| • B.Sc. Computer Science , Bowling Green State University. | Aug 2017 – May 2019 |
| • AWS Certified Machine Learning - Specialty | Sep 2025 |
| • Fundamentals of Accelerated Computing with Modern CUDA C++ , Nvidia. | Mar 2025 |
| • Azure AI Fundamentals , Microsoft. | Mar 2024 |